THE GYSIN TYPE BOUNDARY OF THE LOG COMPLEX IS COMPATIBLE WITH THE MAPS ATTACHED TO THE FIRST CHERN CLASS OF THE DIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The short exact sequence defining the de Rham complex of log poles and its boundary. Same chapter derham.tex, label lemma-log-complex (invoked at proof lines 2568, 2635 and 2672). It gives the shifted sequence $0 \to \Omega^\bullet_{X/S}[1] \to \Omega^\bullet_{X/S}(\log Y)[1] \to \Omega^\bullet_{Y/S} \to 0$ whose boundary is the map δ of the statement.
- The map of the derived category attached to a de Rham class. Same chapter derham.tex, remark remark-cup-product-as-a-map (invoked at proof lines 2573 and 2579). It defines ξ' and ζ' as the maps attached to $c_1^{dR}(\mathcal{O}_X(-Y))$ and to its restriction to Y.
- The punctured line bundle sequence and the identification of its boundary with the Chern class map. Same chapter derham.tex, labels lemma-the-complex-for-L-star (invoked at proof lines 2608 and 2643) and lemma-the-complex-for-L-star-gives-chern-class (invoked at proof lines 2606 and 2641). They describe ξ' and ζ' as boundary maps of the punctured line bundle sequences, which reduces the commutativity to producing morphisms of short exact sequences.

Source reference. Original lemma 0FMW: sources/stacks-derham/pinned-original-source.tex, source lines 2391–2391. Navigation pointer for \ref{lemma-log-complex} used inside the counted span (line 2568); the argument that consumes it is retained in the span.

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Source reference. Original lemma 0FUF: sources/stacks-derham/pinned-original-source.tex, source lines 1348–1348. Navigation pointer for \ref{lemma-the-complex-for-L-star} used inside the counted span (line 2608); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUG: sources/stacks-derham/pinned-original-source.tex, source lines 1422–1422. Navigation pointer for \ref{lemma-the-complex-for-L-star-gives-chern-class} used inside the counted span (line 2606); the argument that consumes it is retained in the span.

Source reference. Original remark 0FU7: sources/stacks-derham/pinned-original-source.tex, source lines 575–575. Navigation pointer for \ref{remark-cup-product-as-a-map} used inside the counted span (line 2573); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $Y \subset X$ be an effective Cartier divisor. Assume the de Rham complex of log poles is defined for $Y \subset X$ over S. Denote*

$$\delta : \Omega^\bullet_{Y/S} \to \Omega^\bullet_{X/S}[2]$$

in $D(X, f^{-1}\mathcal{O}_S)$ the "boundary" map coming from the short exact sequence in Lemma 0FMW. Denote

$$\xi' : \Omega^\bullet_{X/S} \to \Omega^\bullet_{X/S}[2]$$

in $D(X, f^{-1}\mathcal{O}_S)$ the map of Remark 0FU7 corresponding to $\xi = c_1^{dR}(\mathcal{O}_X(-Y))$. Denote

$$\zeta' : \Omega^\bullet_{Y/S} \to \Omega^\bullet_{Y/S}[2]$$

in $D(Y, f|_Y^{-1}\mathcal{O}_S)$ the map of Remark 0FU7 corresponding to $\zeta = c_1^{dR}(\mathcal{O}_X(-Y)|_Y)$. Then the diagram

$$\begin{array}{ccc} \Omega^\bullet_{X/S} & \longrightarrow & \Omega^\bullet_{Y/S} \\ {\scriptstyle \xi'}\downarrow & \swarrow{\scriptstyle \delta} & \downarrow{\scriptstyle \zeta'} \\ \Omega^\bullet_{X/S}[2] & \longrightarrow & \Omega^\bullet_{Y/S}[2] \end{array}$$

is commutative in $D(X, f^{-1}\mathcal{O}_S)$.

Proof. More precisely, we define δ as the boundary map corresponding to the shifted short exact sequence

$$0 \to \Omega^\bullet_{X/S}[1] \to \Omega^\bullet_{X/S}(\log Y)[1] \to \Omega^\bullet_{Y/S} \to 0$$

It suffices to prove each triangle commutes. Set $\mathcal{L} = \mathcal{O}_X(-Y)$. Denote $\pi : L \to X$ the line bundle with $\pi_*\mathcal{O}_L = \bigoplus_{n \geq 0} \mathcal{L}^{\otimes n}$.

Commutativity of the upper left triangle. By Lemma 0FUG the map ξ' is the boundary map of the triangle given in Lemma 0FUF. By functoriality it suffices to prove there exists a morphism of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Omega^\bullet_{X/S}[1] & \longrightarrow & \Omega^\bullet_{L^\star/S,0}[1] & \longrightarrow & \Omega^\bullet_{X/S} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \Omega^\bullet_{X/S}[1] & \longrightarrow & \Omega^\bullet_{X/S}(\log Y)[1] & \longrightarrow & \Omega^\bullet_{Y/S} & \longrightarrow & 0 \end{array}$$

where the left and right vertical arrows are the obvious ones. We can define the middle vertical arrow by the rule

$$\omega' + \mathrm{d}\log(s) \wedge \omega \longmapsto \omega' + \mathrm{d}\log(f) \wedge \omega$$

where ω', ω are local sections of $\Omega^\bullet_{X/S}$ and where s is a local generator of $\mathcal{L}$ and $f \in \mathcal{O}_X(-Y)$ is the corresponding section of the ideal sheaf of Y in X. Since the constructions of the maps in Lemmas 0FUF and 0FMW match exactly, this works.

Commutativity of the lower right triangle. Denote $\overline{L}$ the restriction of L to Y. By Lemma 0FUG the map ζ' is the boundary map of the triangle given in Lemma 0FUF using the line bundle $\overline{L}$ on Y. By functoriality it suffices to prove there exists a morphism of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Omega^\bullet_{X/S}[1] & \longrightarrow & \Omega^\bullet_{X/S}(\log Y)[1] & \longrightarrow & \Omega^\bullet_{Y/S} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \Omega^\bullet_{Y/S}[1] & \longrightarrow & \Omega^\bullet_{\overline{L}^\star/S,0}[1] & \longrightarrow & \Omega^\bullet_{Y/S} & \longrightarrow & 0 \end{array}$$

where the left and right vertical arrows are the obvious ones. We can define the middle vertical arrow by the rule

$$\omega' + \mathrm{d}\log(f) \wedge \omega \longmapsto \omega'|_Y + \mathrm{d}\log(\overline{s}) \wedge \omega|_Y$$

where ω', ω are local sections of $\Omega^\bullet_{X/S}$ and where f is a local generator of $\mathcal{O}_X(-Y)$ viewed as a function on X and where $\overline{s}$ is $f|_Y$ viewed as a section of $\mathcal{L}|_Y = \mathcal{O}_X(-Y)|_Y$. Since the constructions of the maps in Lemmas 0FUF and 0FMW match exactly, this works. □